

1. Title

Scientific Collaboration Network Analyzer

2. Objective

Develop a research collaboration management platform that enables universities and research organizations to manage publications, researchers, institutions, projects, conferences, and collaborations.

The system tracks co-authorship, research groups, publications, funding projects, conference participation, citations, and institutional partnerships through a centralized database.

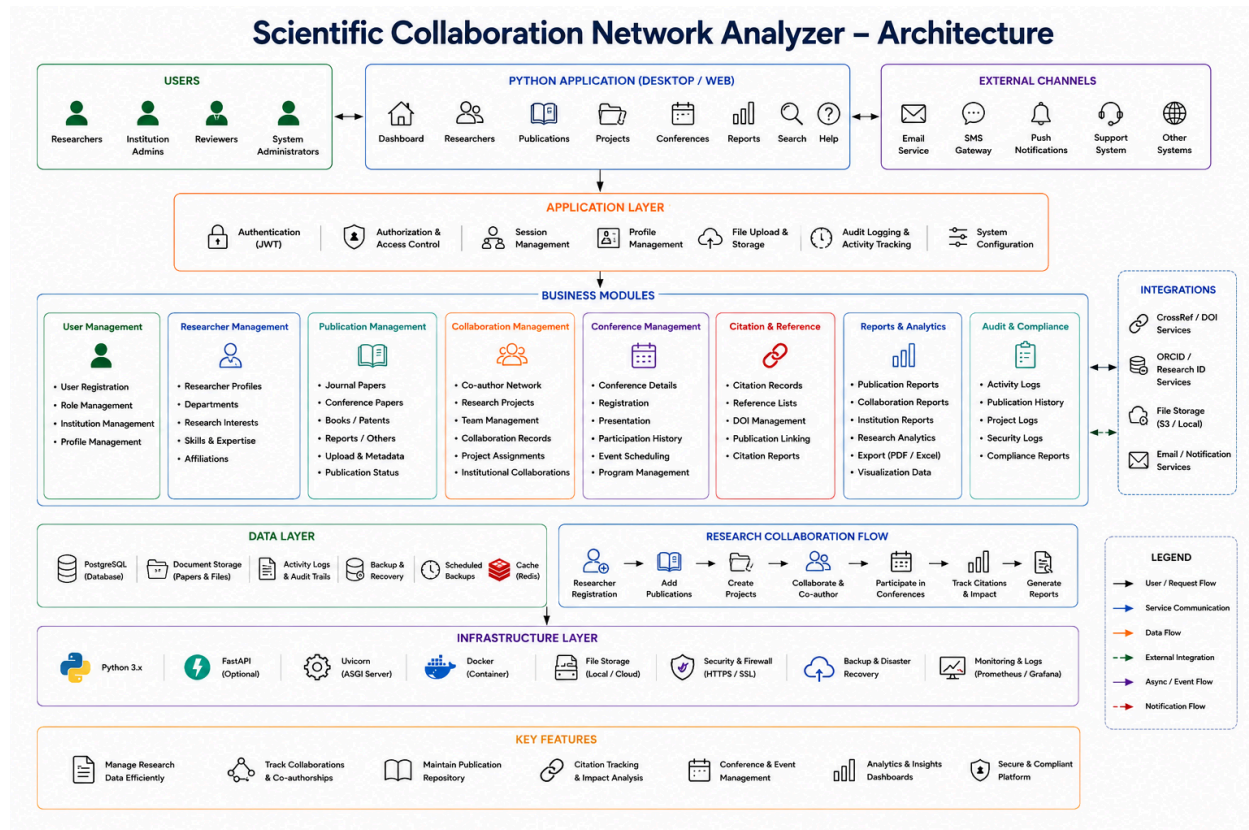
It provides visualization of collaboration networks, publication statistics, institutional reports, and researcher profiles without using AI-based analysis.

The platform is suitable for universities, research institutes, government laboratories, academic publishers, and funding organizations.

Outcomes

- (i) Develop a research management system.
 - (ii) Implement publication repository.
 - (iii) Build collaboration network management.
 - (iv) Develop conference and project tracking.
 - (v) Build institutional collaboration workflows.
 - (vi) Generate publication analytics.
 - (vii) Develop reporting dashboards.
 - (viii) Deploy using Docker.
-

3. Architecture Components



4. Modules

1. User Management

Roles

- Researcher
- Institution Admin
- Reviewer
- System Admin

Features

- Login
- Registration
- Researcher Profile
- Institution Management

2. Researcher Management

- Academic profiles
- Departments
- Skills
- Research interests
- Affiliations

3. Publication Management

- Journal papers
- Conference papers
- Books
- Patents
- Technical reports
- Publication uploads

Publication Status

- Draft
- Submitted
- Published
- Archived

4. Collaboration Management

- Co-author records
- Research projects
- Institutional collaborations
- Team management
- Project assignments

5. Conference Management

- Conference registration

- Presentation records
 - Participation history
 - Event scheduling
-

6. Citation & Reference Module

- Citation records
 - Reference lists
 - Publication linking
 - DOI management
-

7. Dashboard

Researcher Dashboard

- Publications
- Projects
- Conferences
- Collaborators

Institution Dashboard

- Departments
- Publications
- Active Projects
- Collaboration Statistics

Admin Dashboard

- Overall Reports
 - Institution Analytics
 - User Statistics
-

8. Reports & Export

- Publication Reports
- Research Reports
- Collaboration Reports

- Institution Reports
 - Excel Export
 - PDF Export
-

9. Audit Module

- User activity
 - Publication history
 - Project logs
 - Security logs
-

5. Week-wise Timeline

Milestone 1 (Week 1-2)

Tasks

- Requirement gathering
- Database schema
- FastAPI setup
- React setup
- Authentication
- Researcher profiles

Outcomes

- Authentication completed
 - User management functional
 - Researcher module completed
-

Milestone 2 (Week 3-4)

Tasks

- Publication module

- Conference module
- Institution management
- File uploads

Outcomes

- Publication repository completed
 - Conference workflows implemented
 - Institution management operational
-

Milestone 3 (Week 5-6)

Tasks

- Collaboration module
- Citation management
- Reports
- Dashboards
- Notifications

Outcomes

- Collaboration tracking completed
 - Reporting module functional
 - Dashboards operational
-

Milestone 4 (Week 7-8)

Tasks

- Testing
- Performance optimization
- Docker deployment
- Documentation
- Final presentation

Outcomes

- Stable production deployment

- Complete documentation
 - Successful demonstration
-

6. Evaluation Criteria

Week 2

- Authentication completed
- Researcher profiles working
- Database implemented

Week 4

- Publication management completed
- Conference module working
- Institution workflows functional

Week 6

- Collaboration tracking completed
- Reports generated
- Dashboard operational

Week 8

- Testing completed
 - Deployment completed
 - Documentation submitted
-

7. Tools & Tech Stack

Backend

- Python
- FastAPI

Frontend

- Tkinter
- PyQt
- Flask

Database

- PostgreSQL
- Redis

Authentication

- JWT
- OAuth2

Backend Libraries

- SQLAlchemy
- Alembic
- Pydantic
- Uvicorn

Storage

- AWS S3 / Local Storage

DevOps

- Docker
- GitHub
- GitHub Actions
- Postman

8. Performance Metrics

Publication Management

- Publication upload success rate
- Collaboration creation rate
- Report generation time

System Performance

- API response time
- Concurrent users
- Database query performance

Research Management

- Active researchers
- Publications tracked
- Projects managed
- Collaboration records maintained